

Amir (Ghasem) Boshrouei Shargh

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Ph.D. researcher developing machine learning and uncertainty-aware computational methods for engineering design, with expertise in active learning, generative modeling, structural reliability, and seismic performance assessment.

Education

PhD in Structural Engineering

The Pennsylvania State University, PA, USA

Jan. 2023 – Present

CGPA (WES): 3.8/4

NHERI Converging Design Project

Multi-objective Optimization of Resilient Structural Spines

ICDS Mid-Scale Seed Grant

Reinforcement Learning for High-Dimensional Adaptive Surrogate Modeling

Relevant Courses:

Probabilistic Structural Reliability Analysis, Computational Intelligence and Analysis of Randomness, Stochastic Processes and Monte Carlo Methods, Pattern Recognition and Machine Learning, Structural Design for Dynamic Loads, Computational Design and Optimization.

MSc in Earthquake Engineering

Ferdowsi University of Mashhad, Iran

Sep. 2015 – Sep. 2018

CGPA (WES): 3.88/4

(Full Scholarship)

Thesis Title: *Seismic Assessment of Structures Considering Spectral Shape Parameters*

Relevant Courses:

Random Vibrations, Seismic Structural Design, Probabilistic Seismic Hazard Analysis, Advanced Seismology, Structural Rehabilitation and Retrofitting, Structural Dynamics, Theory of Elasticity and Plasticity.

BSc in Civil & Environmental Engineering

Shahrood University of Technology, Iran

Sep. 2011 – Jul. 2015

(Full Scholarship)

Publications

Journal Articles

- Shargh, G. B., Papakonstantinou, K. G., Brown, N. (2026) “Generative Modeling for Inverse Engineering Design.” In preparation.
- Shargh, G. B., Barbosa, A. R., van de Lindt, J. W., Brown, N. C. (2026). “A Parametric Computational Framework for Modeling and Seismic Design of Post-Tensioned Mass Timber Rocking Walls.” *Earthquake Engineering & Structural Dynamics*. Accepted for publication.
- Shargh, G. B., Papakonstantinou, K. G., van de Lindt, J. W., Barbosa, A. R., Simpson, B., Sinha, A., Brown, N. (2026) “Reliability-Based Design Optimization of Mass Timber Self-Centering Rocking Walls.” *Journal of Structural Engineering*. DOI: [10.1061/JSENDH.STENG-15751](https://doi.org/10.1061/JSENDH.STENG-15751)
- Shargh, G. B., Barati, R. (2023) “Probabilistic Assessment of Ground Motion Rotation Impact on Seismic Performance of Structures.” *Engineering Structures*. DOI: [10.1016/j.engstruct.2023.116383](https://doi.org/10.1016/j.engstruct.2023.116383)
- Shargh, G. B., Barati, R. (2021) “Estimation of Inelastic Seismic Input Energy.” *Soil Dynamics and Earthquake Engineering*. DOI: [10.1016/j.soildyn.2020.106505](https://doi.org/10.1016/j.soildyn.2020.106505)

Conference Papers

- **Shargh, G. B.**, Brown, N. (2026) “*Physics-Aware Bayesian Optimization for Reliable PT-CLT Rocking Walls: Impact of Feature Representations.*” 13th U.S. National Conference on Earthquake Engineering (13NCEE), EERI.
- **Shargh, G. B.**, Barbosa, A., Simpson, B., Sinha, A., van de Lindt, J. W., Brown, N. (2024) “*Multi-Objective Optimization of Cross-Laminated Timber Shear Walls.*” 18th World Conference on Earthquake Engineering.
- **Shargh, G. B.**, Barati, R., Brown, N. (2024) “*Angular Efficiency of Intensity Measures.*” 18th World Conference on Earthquake Engineering.
- Barati, R., **Shargh, G. B.** (2024) “*Seismic Intensity Measure Selection Considering Record-to-Record and Angle-to-Angle Uncertainties.*” 9th International Conference on Seismology and Earthquake Engineering.
- Barati, R., **Shargh, G. B.**, Karamodin, A. (2019) “*Evaluation of Methods Estimating Structural Strength Reduction Factor under Far-Field Records.*” 8th International Conference on Seismology & Earthquake Engineering.

Open-Source Software

- [Parametric-PT-CLT](#) — A parametric finite element modeling framework for post-tensioned cross-laminated timber (PT-CLT) rocking wall systems. Automates model generation, seismic design, and nonlinear analysis in OpenSeesPy while supporting seismic performance assessment, risk and reliability analysis, and performance-based design optimization.

Professional Experience

Research Assistant, The Pennsylvania State University Jan. 2023 – Present

- Developed an active-learning framework for generative inverse design in high-dimensional spaces.
- Developed Python-based parametric FEM and probabilistic models for PT-CLT rocking walls.
- Developed ML surrogate models for efficient structural optimization and uncertainty quantification.
- Applied physics-aware Bayesian optimization for reliability-based design.
- Performed sensitivity, reliability, and seismic performance analyses of resilient structural systems.

Research Assistant, Ferdowsi University of Mashhad Oct. 2018 – Jan. 2023

- Conducted probabilistic seismic analysis and nonlinear dynamic simulations using OpenSees.

Awards and Honors

- Mid-Scale Seed Grant, Institute for Computational and Data Sciences (ICDS), The Pennsylvania State University.
- Full Research Assistantship (RA), NSF-funded research project.
- Full national scholarships for graduate and undergraduate studies.
- Top 5%, M.Sc. graduating class, Ferdowsi University of Mashhad.
- Top 4%, Iranian M.Sc. entrance examination in Civil Engineering.
- Top 1%, Iranian national university entrance examination.
- 6th Place, Khwarizmi National RoboCup Student Competition.

Technical Skills

Programming & ML: Python (NumPy, SciPy, TensorFlow, PyTorch), MATLAB, R, C++, Fortran

ML & Data Methods: Active Learning, Generative Modeling, Surrogate Modeling, Bayesian Optimization, Reinforcement Learning, Uncertainty Quantification

Engineering & Simulation: OpenSeesPy, OpenSees, ETABS, SAP2000, AutoCAD

Structural & Probabilistic Methods: Reliability-Based Design, Seismic Risk Assessment, Probabilistic Seismic Assessment, Sensitivity Analysis